DeFi Token Sustainability Report

Executive Summary

This report delivers a concise, data-driven assessment of long-term token sustainability for six core DeFi protocols using live CoinGecko and DefiLlama data. We calculate five critical metrics - annualized TVL growth, future dilution potential from supply schedules, emission pressure relative to growth, P/S ratio based on real fee revenue, and FDV/Revenue - then combine them into a single composite risk score (lower = more sustainable). Emission Pressure is a heuristic metric: when TVL growth is positive, it equals supply_overhang / TVL_growth (floored at 5% growth); when negative, it equals (supply_overhang x 2) + (|decline| x 2). Both branches are capped at 10 to avoid extremes. It measures how much undistributed supply pressure is absorbed or amplified by protocol momentum. Standard composite weights (protocols where revenue accrues to token holders): Future Dilution Potential 25%, Emission Pressure 20%, Valuation Risk (P/S) 20%, TVL Instability 20%, FDV/Revenue 15%. Adjusted weights (governance-only tokens where protocol revenue does not flow to holders, e.g. Ethena, Lido): Future Dilution Potential 30%, Emission Pressure 25%, Valuation Risk (P/S) 10%, TVL Instability 20%, FDV/Revenue 15%. Note: Composite scores are relative to this peer group; values will shift with different protocol selections.

Protocols Analyzed & Why They Were Chosen

Pendle (PENDLE) - the leading yield-tokenization protocol with $2.2B TVL - was chosen because it pioneered PT/YT splitting for fixed and leveraged yield trading. High emissions for liquidity incentives (0.4123 supply overhang ratio) but strong revenue capture make it the perfect example of 2026 DeFi primitives. Note: TVL peaked near $7B in mid-2024 driven by Eigenlayer and Ethena points-farming programs; the current figure reflects organic demand after those incentive cycles concluded.

Ethena (ENA) - the pioneer of synthetic dollar (USDe) and delta-neutral yield with $6.7B+ TVL - was included as the breakout stablecoin innovation of 2025-2026. With 0.4339 supply overhang ratio and +23.20% TVL growth, it shows funding-rate mechanics creating sustainable yield.

GMX (GMX) - the benchmark on-chain perpetuals DEX with fee-sharing to stakers (note: V1 distributed 30% to GMX stakers; V2 revised the split across GLP/GM pools) - was selected to represent real revenue-aligned tokenomics. With controlled inflation and proven holder value accrual, it contrasts emission-heavy models despite -38.75% TVL growth.

Lido (LDO) - the dominant liquid staking protocol with $18.3B+ TVL - was chosen as the leading example of staking innovation. With 0.1507 supply overhang ratio and -8.60% TVL growth, it shows how real utility drives sustainable token value.

Curve Finance (CRV) - the premier stablecoin DEX with its iconic veCRV locking model - highlights advanced tokenomics in competitive markets. Its 0.5105 supply overhang ratio and -2.32% growth illustrate the trade-offs of emission-heavy designs.

Sky (SKY, formerly MakerDAO) - the original decentralized stablecoin issuer - represents mature tokenomics with buybacks and collateral interest. Despite -15.28% TVL growth and 0.0181 future dilution potential, it remains a benchmark for long-term holder alignment. Note: supply metrics reflect the post-rebrand state (1 MKR = 24,000 SKY conversion), which substantially inflates circulating supply figures and should be interpreted in that context.

What the Final Three Charts Reveal

- Relative Dilution chart shows exactly what percentage of each token's maximum supply will unlock over the next

3 years - revealing future selling pressure on holders. Note: Assumes linear unlocks for simplicity; actual vesting may be non-linear or emissions-based.

- 3-Year Circulating Supply Projection displays absolute token growth in millions, allowing direct visual comparison of emission scale across protocols.
- Sustainability Composite Score Ranking bar chart synthesizes all five metrics into one final score. Sky exhibits the lowest score at 0.083, indicating the strongest sustainability among the six.

All data pulled March 2026 - One-click Colab - Ready for investor decks

Metric Definitions

Reference guide to every metric shown in the protocol tables.

Market Cap

Total market value of all tokens currently in circulation (price x circulating supply). Low: smaller protocol by market valuation. High: larger, more established market presence.

TVL (Total Value Locked)

Total assets actively deposited and utilized within the protocol. Reflects real capital trust and usage. Low: limited capital engagement or early-stage protocol. High: strong user adoption and capital depth.

TVL Growth (ann.)

Annualized CAGR of TVL over the trailing 12 months. Measures whether the protocol is growing or shrinking. Negative: capital or users are exiting. Positive: growing adoption and inflows.

TVL Instability

Derived from TVL Growth: max(0, -TVL_growth). Protocols with positive TVL growth score 0 on this dimension (no instability penalty), while declining protocols are penalized proportionally to their rate of contraction. This means a fast-growing protocol like Ethena receives a perfect 0 here despite other risk factors.

Future Dilution Potential

Proportion of the maximum token supply not yet in circulation: (max_supply - circ_supply) / max_supply. Scale: 0 to 1. Low (near 0): most supply is already circulating - minimal future unlock pressure on holders. High (near 1): large proportion of supply still to be released - greater future sell pressure risk. Note: assumes all remaining supply will eventually circulate. For protocols with continuous emission schedules (e.g. CRV), actual release rate may differ substantially from what this figure implies.

Emission Pressure

Heuristic metric combining supply overhang (undistributed supply as a fraction of max supply) with TVL momentum as a relative ranking signal. Not derived from standard financial theory. Positive TVL growth: overhang / TVL_growth (floored at 5% to prevent near-zero growth from creating extreme scores). Negative TVL growth: (overhang x 2) + (|TVL decline| x 2). Both branches capped at 10. Low: unlock pressure is well-absorbed by growth momentum. High: significant supply overhang against weak or declining protocol traction. Scores are only meaningful relative to this peer group.

P/S Ratio (Price-to-Sales)

Market Cap divided by annualized protocol revenue (30-day average daily revenue x 365). Low: token is inexpensive relative to current revenue generation - potentially undervalued. High: token trades at a revenue premium - market expects significant future growth. Governance-only tokens (e.g. ENA, LDO) where revenue flows to the protocol treasury rather than token holders are assigned half the standard P/S weight in the composite score, as market cap / protocol revenue is not a direct holder value metric in those cases.

FDV / Revenue

Fully Diluted Valuation (price x max supply) divided by annualized revenue. A forward-looking valuation check. Low: reasonable forward valuation relative to current earnings. High: protocol is priced for aggressive future revenue growth well beyond current levels.

Composite Score

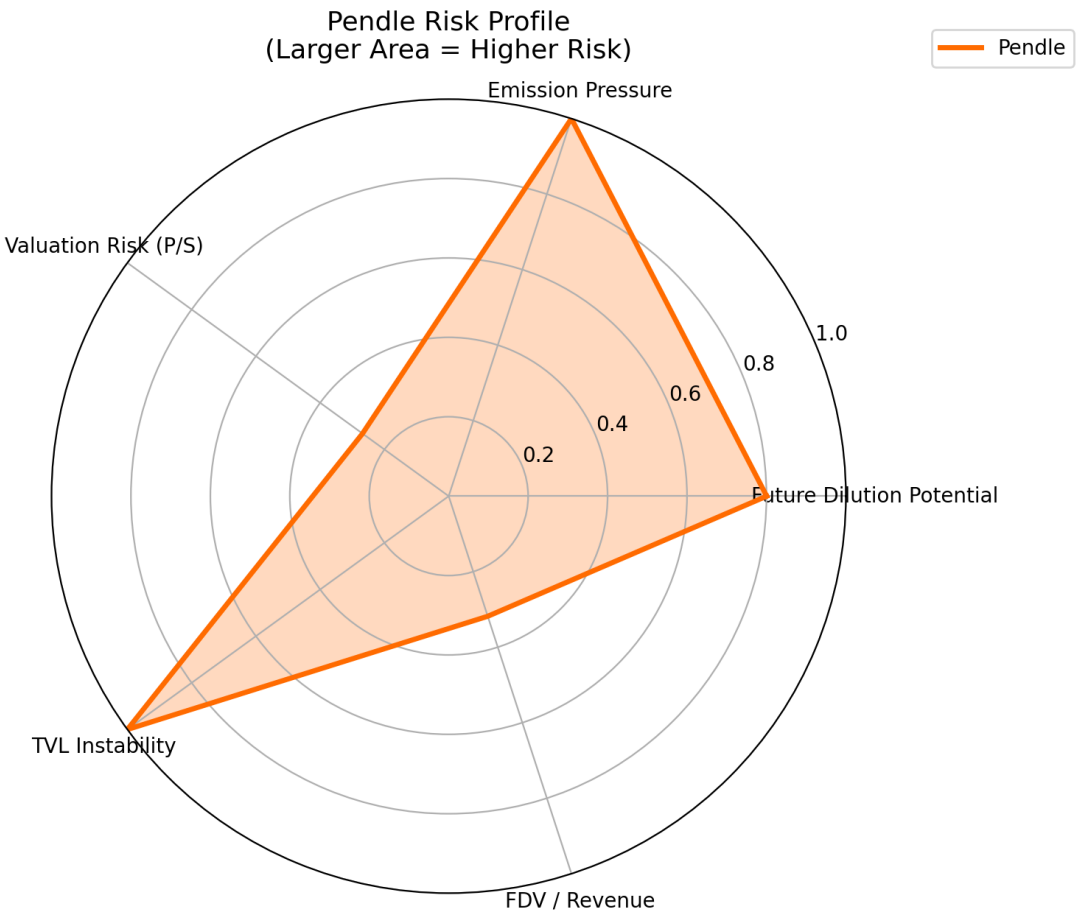
Weighted sum of all five risk metrics, each min-max normalized within this peer group (0-1 scale). Standard weights (protocols where revenue accrues to token holders): Future Dilution Potential 25% | Emission Pressure 20% | Valuation Risk (P/S) 20% | TVL Instability 20% | FDV/Revenue 15%. Adjusted weights (protocols where protocol revenue does not flow to token holders, e.g. Ethena, Lido): Future Dilution Potential 30% | Emission Pressure 25% | Valuation Risk (P/S) 10% | TVL Instability 20% | FDV/Revenue 15%. The P/S weight is halved for governance-only tokens because market cap / protocol revenue is not a holder value metric in those cases. Implementation note: P/S, FDV/Revenue, and Emission Pressure are log-scaled (log1p) before min-max normalization to reduce distortion from extreme outliers. LOWER score = more sustainable tokenomics. Scores are relative to this peer group and will shift if the protocol selection changes.

Pendle

Metric	Value
Market Cap	$201,709,891
TVL	$2,151,786,849
TVL as of	2026-03-07
Future Dilution Potential	0.4123
TVL Growth (ann.)	-52.59%
Emission Pressure	1.876
P/S Ratio	18.86
FDV / Revenue	32.12
Composite Score	0.7015

AI Analyst Insight

Pendle maintains a substantial TVL of over $2.15 billion. However, an Emission Pressure of 1.876 indicates emissions are outpacing TVL growth absorption capacity, reflecting significant token inflation. This results in a Composite Score of 0.7015, placing Pendle among the 2nd least sustainable protocols in its peer group (rank 5 of 6).

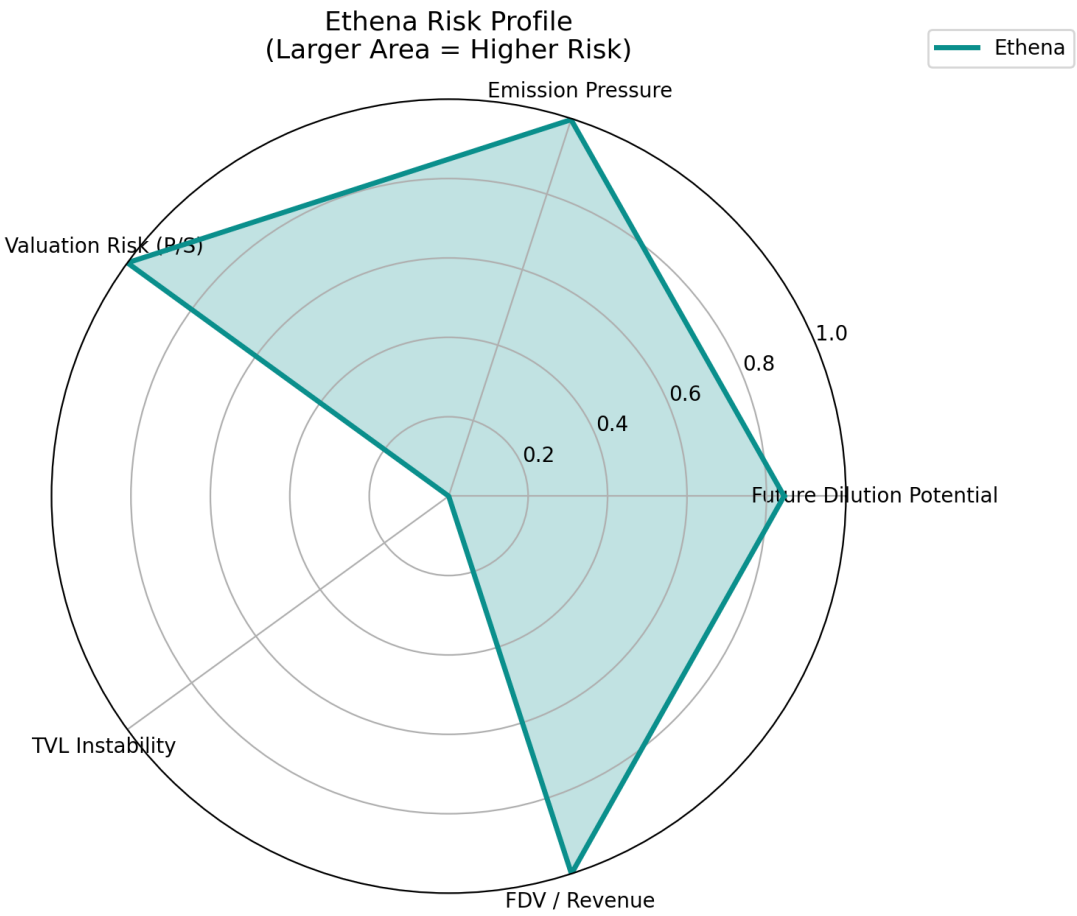


Ethena

Metric	Value
Market Cap	$863,361,472
TVL	$6,692,517,712
TVL as of	2026-03-07
Future Dilution Potential	0.4339
TVL Growth (ann.)	+23.20%
Emission Pressure	1.870
P/S Ratio	553.18
FDV / Revenue	977.11
Composite Score	0.7526

AI Analyst Insight

Ethena exhibits substantial platform scale with a Total Value Locked (TVL) of $6.69 billion. However, its Emission Pressure of 1.870 indicates emissions outpacing TVL growth absorption capacity. This results in a Composite Score of 0.7526, placing the protocol among the least sustainable in its peer group with a rank of 6 of 6.

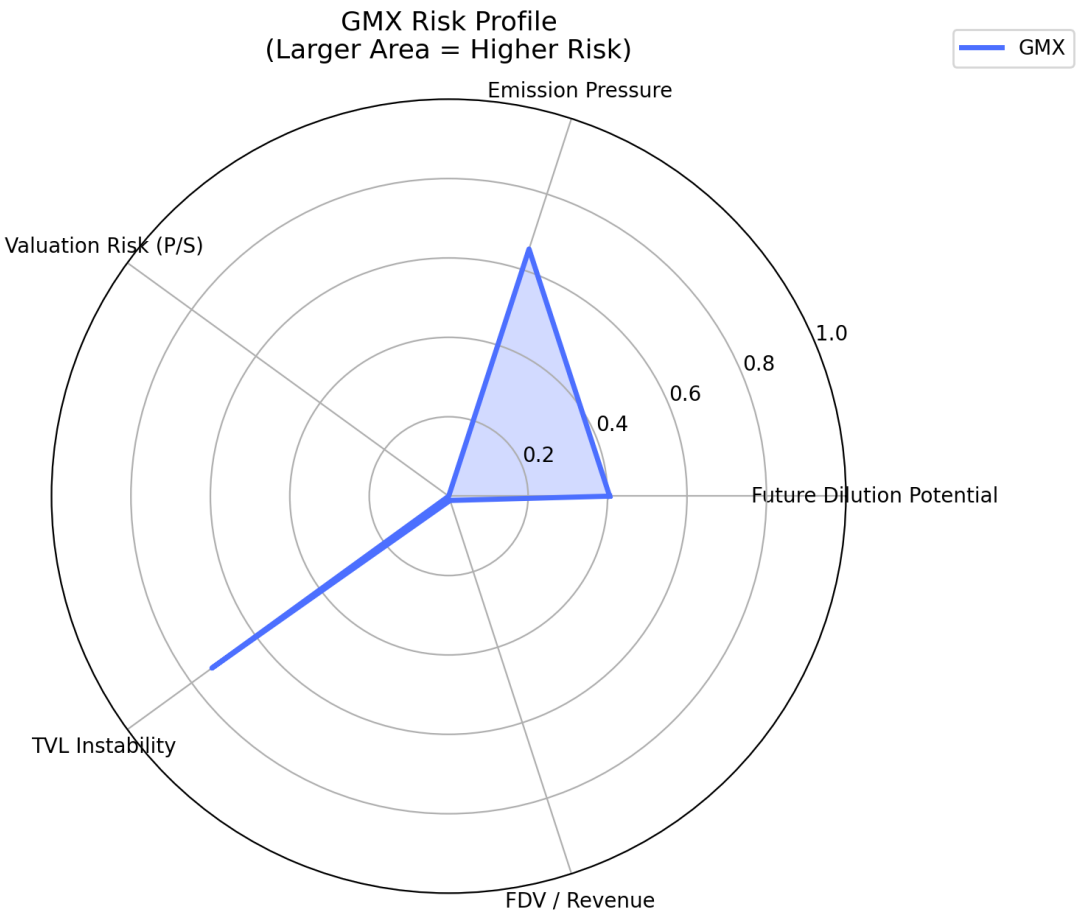


GMX

Metric	Value
Market Cap	$70,649,198
TVL	$263,563,231
TVL as of	2026-03-07
Future Dilution Potential	0.2176
TVL Growth (ann.)	-38.75%
Emission Pressure	1.210
P/S Ratio	4.87
FDV / Revenue	6.22
Composite Score	0.3813

AI Analyst Insight

GMX exhibits a favorable valuation with an FDV/Revenue multiple of 6.22. However, an Emission Pressure of 1.210 indicates that token emissions are outpacing TVL growth absorption capacity. GMX holds a Composite Score of 0.3813, placing this protocol in the middle of its peer group (rank 3 of 6).

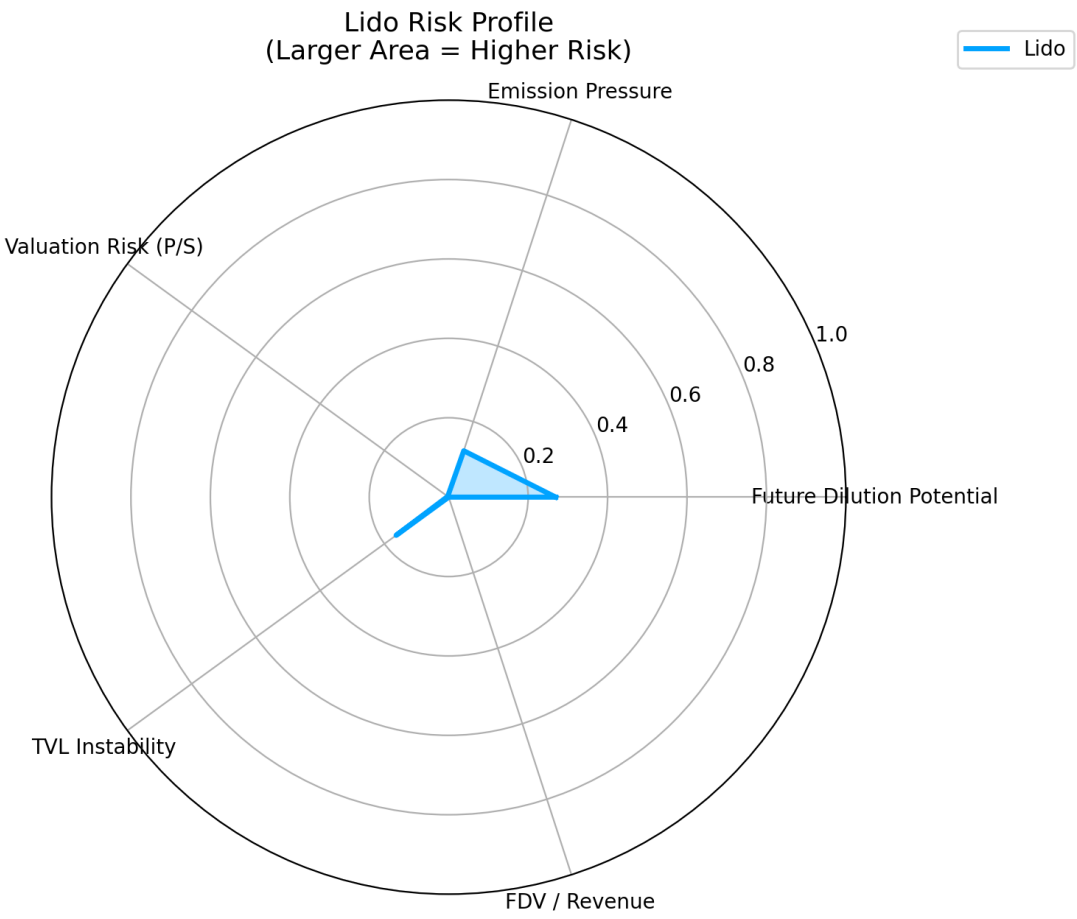


Lido

Metric	Value
Market Cap	$248,579,909
TVL	$18,258,513,501
TVL as of	2026-03-07
Future Dilution Potential	0.1507
TVL Growth (ann.)	-8.60%
Emission Pressure	0.473
P/S Ratio	4.95
FDV / Revenue	5.82
Composite Score	0.1446

AI Analyst Insight

Lido's P/S ratio of 4.95 reflects protocol-level earnings that flow to the DAO treasury; LDO token holders have no direct revenue claim, unlike GMX stakers or veCRV lockers. A -8.6% TVL growth indicates recent deceleration, posing a short-term absorption challenge for new emissions. Lido holds a Composite Score of 0.1446, positioning it as the 2nd most sustainable (rank 2 of 6) among its peer group.

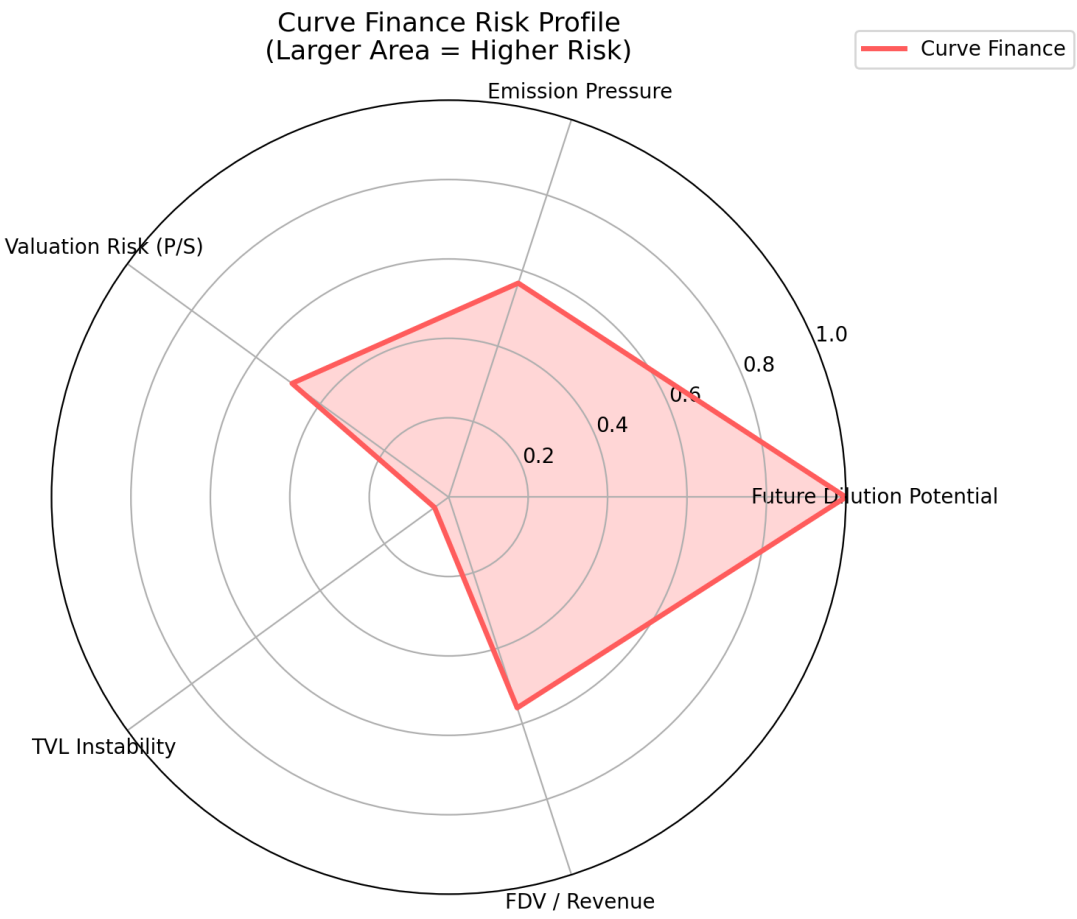


Curve Finance

Metric	Value
Market Cap	$350,125,611
TVL	$1,761,580,061
TVL as of	2026-03-07
Future Dilution Potential	0.5105
TVL Growth (ann.)	-2.32%
Emission Pressure	1.067
P/S Ratio	52.87
FDV / Revenue	108.02
Composite Score	0.5534

AI Analyst Insight

Curve Finance maintains a substantial Total Value Locked (TVL) of $1,761,580,061, indicating significant platform utilization. However, an Emission Pressure of 1.067 indicates emissions outpacing TVL growth absorption capacity. This results in a Composite Score of 0.5534, which places the protocol in the middle of its peer group (rank 4 of 6) regarding long-term token sustainability.

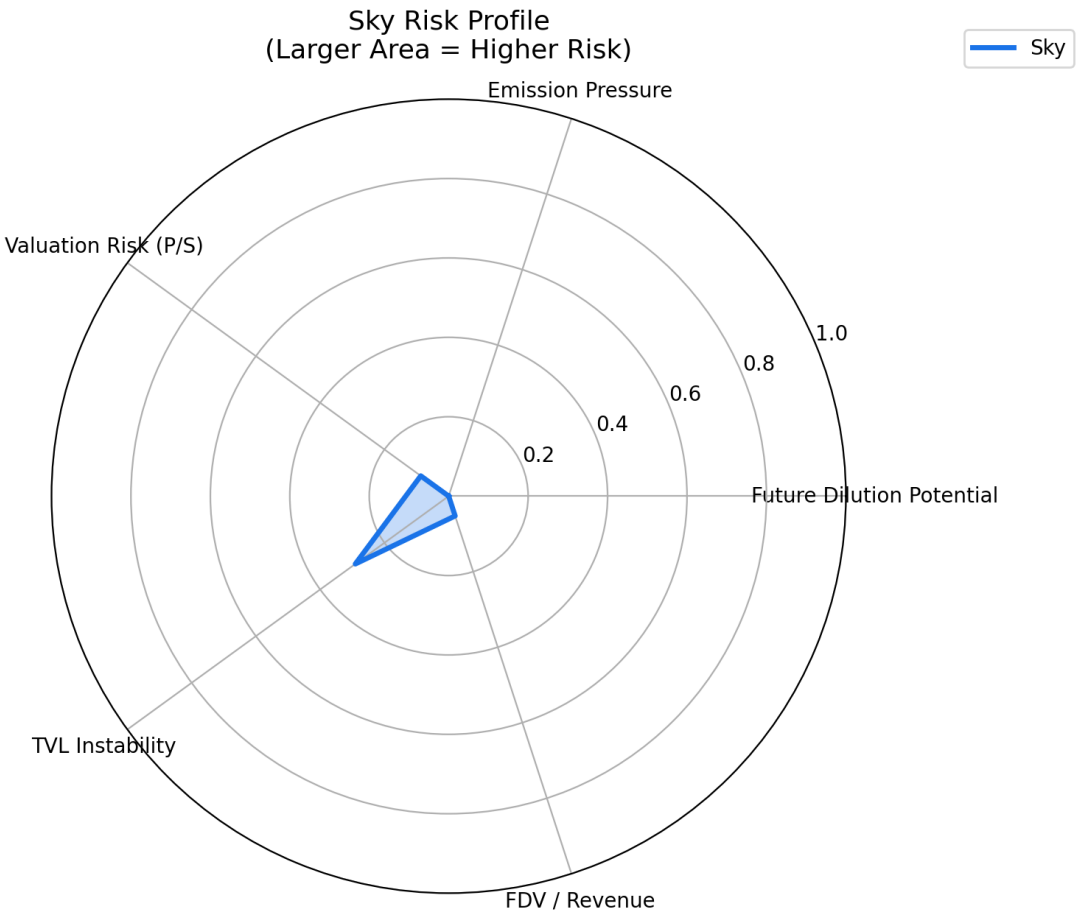


Sky

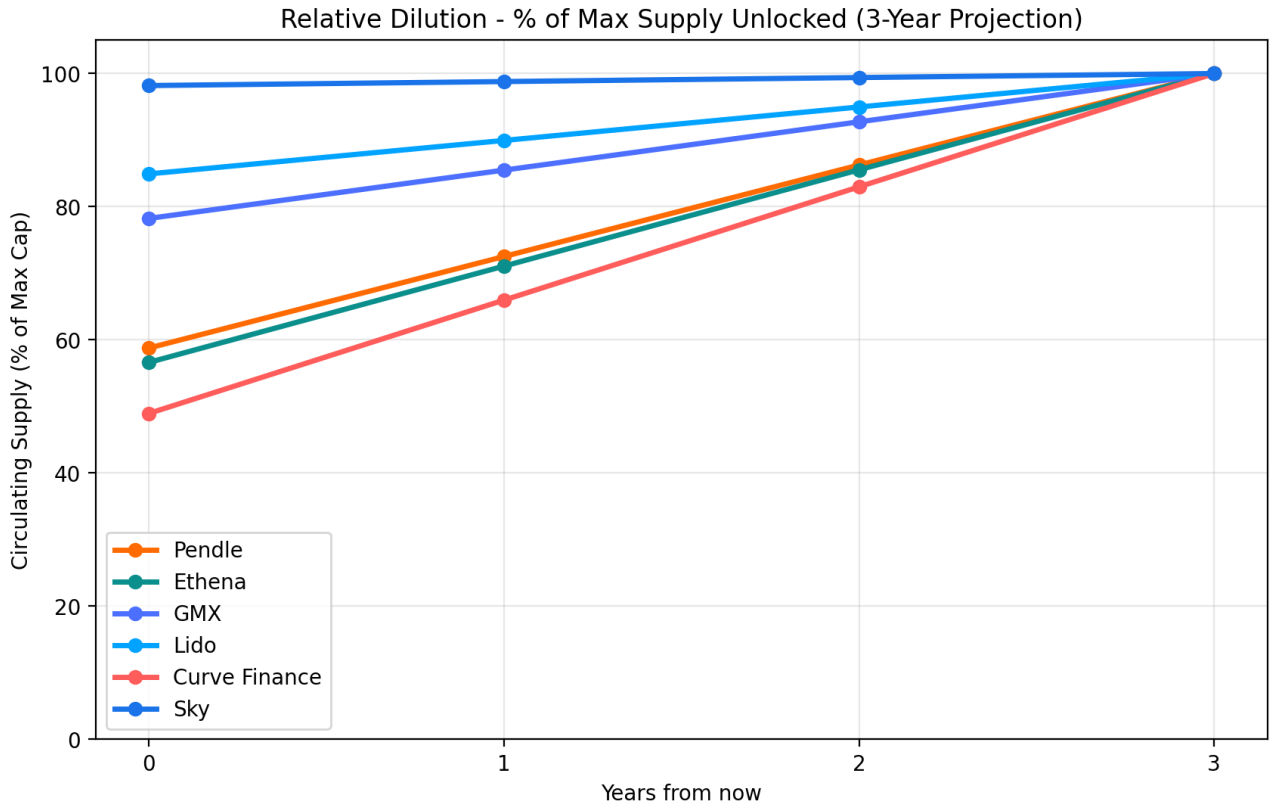
Metric	Value
Market Cap	$1,667,742,643
TVL	$6,873,527,240
TVL as of	2026-03-07
Future Dilution Potential	0.0181
TVL Growth (ann.)	-15.28%
Emission Pressure	0.342
P/S Ratio	7.71
FDV / Revenue	7.85
Composite Score	0.0834

AI Analyst Insight

Sky exhibits a very low future dilution potential of 0.0181, indicating minimal overhang from undistributed token supply. However, its TVL growth is contracting at -15.3%, suggesting a decline in protocol capital absorption. Sky holds a Composite Score of 0.0834, positioning it as the most sustainable (rank 1) among its 6-protocol peer group.



Relative Dilution - % of Max Supply (3-Year)

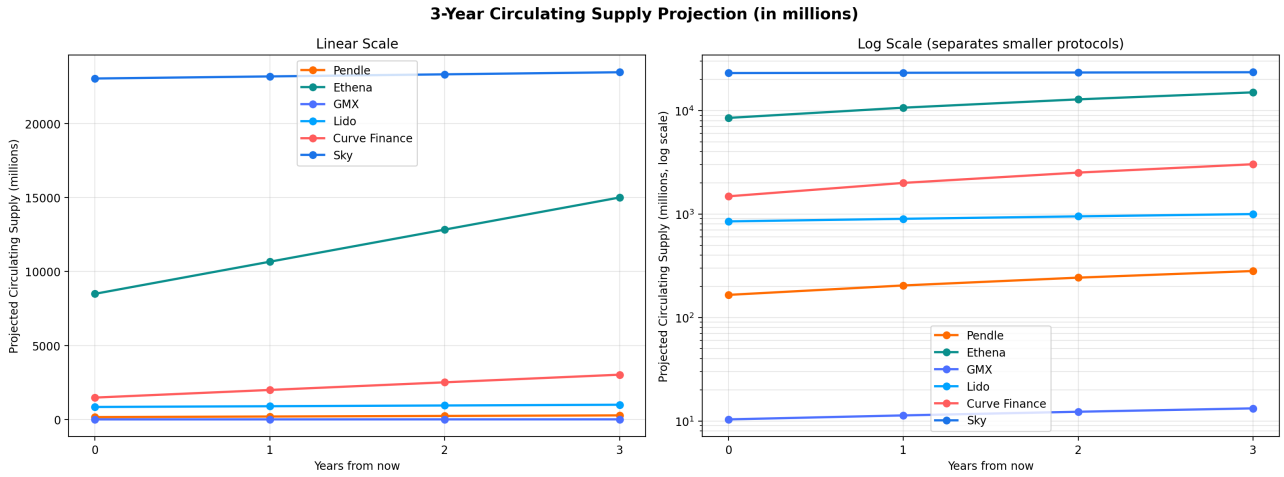


Methodology note: Projections assume linear unlocks across 3 years. This is a simplification - actual schedules vary significantly. CRV emissions follow an exponential decay curve per the Curve whitepaper, so its 104% figure is an overestimate of realistic near-term dilution. Lido's LDO has no formal remaining unlock schedule; its figure reflects undistributed supply only.

AI Analyst Insight

The 3-year supply increase percentage, relative to the initial circulating supply, is a critical metric for token holders, quantifying the potential for future dilution and associated long-term selling pressure. Sky stands out with the lowest supply expansion at 2%, contrasting sharply with Curve Finance at 104% and Ethena at 77%, which show the highest issuance rates. This wide divergence indicates that tokens with high supply growth will inherently experience greater selling pressure, impacting value retention for long-term holders compared to those with minimal new supply. Lido (18%), GMX (28%), and Pendle (70%) fall in the middle range, representing moderate inflation that requires balanced assessment against their utility and demand.

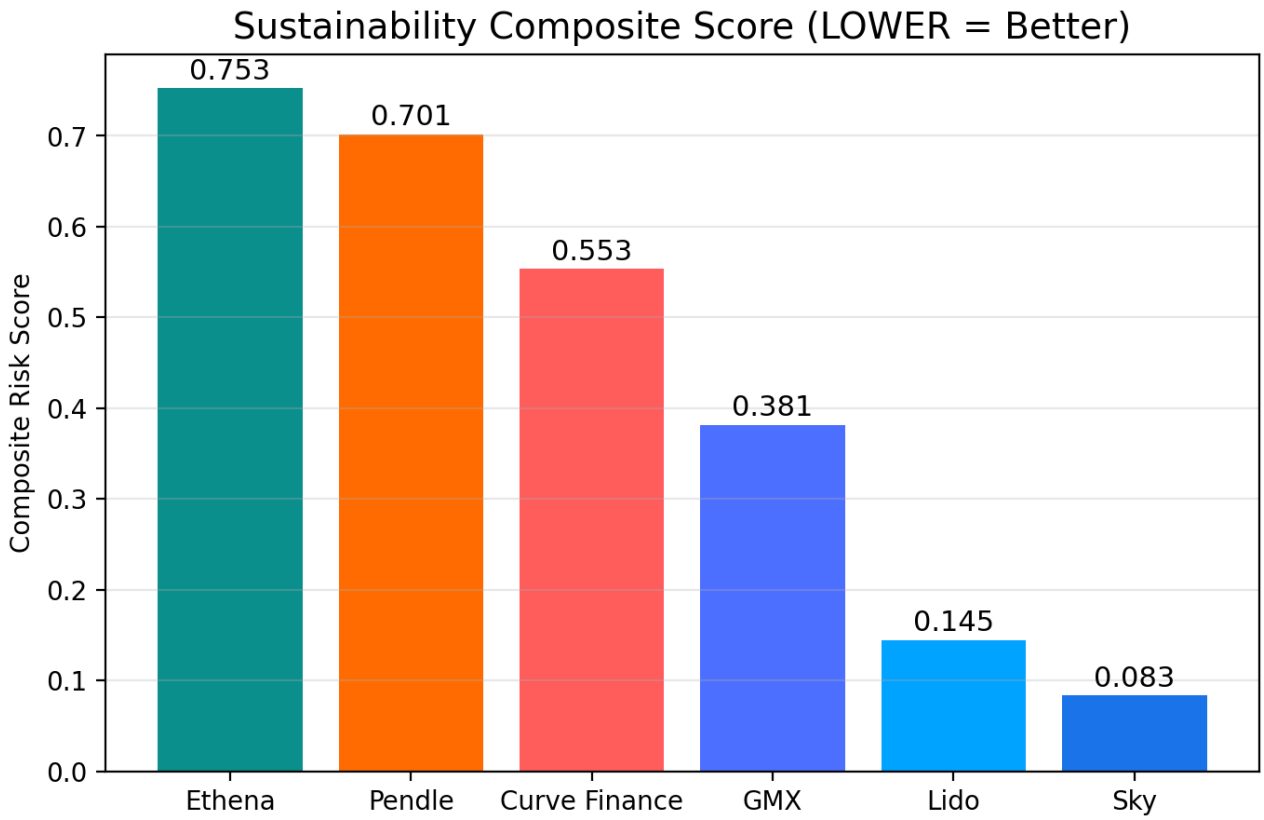
3-Year Circulating Supply Projection (in millions)



AI Analyst Insight

Token dilution directly impacts long-term price sustainability for holders by quantifying the rate at which new supply enters circulation relative to existing supply, which this chart illustrates over a three-year projection. Sky (2%) and Lido (18%) present the lowest projected three-year dilution rates, whereas Ethena (77%) and Curve Finance (104%) represent the highest, indicating Curve's supply is expected to more than double. This significant spread implies that tokens with high projected dilution rates may face greater downward price pressure unless demand growth substantially outpaces supply expansion, potentially impacting long-term holding strategies. GMX (28%) and Pendle (70%) reside in the mid-range, positioning them between the minimal dilution projected for Sky/Lido and the substantial inflation observed in Ethena/Curve.

Sustainability Composite Score Ranking



AI Analyst Insight

This metric quantifies the sustainability profile of a DeFi protocol's tokenomics, providing insight into potential long-term value retention for token holders by assessing underlying economic health. The range highlights Sky at 0.083 as the most sustainable, while Ethena stands at the opposite end with 0.753, indicating the least sustainable profile among the group. This significant spread suggests that protocols with higher scores may face greater challenges in maintaining token value, potentially prompting re-evaluation of holding strategies, whereas lower-scoring protocols might present more resilient investment opportunities. GMX (0.381) and Curve Finance (0.553) represent the middle tier, exhibiting profiles that are notably more sustainable than Ethena but less so than Sky or Lido.